PUKAAR

Pukaar: a call, a cry for help.

TRACK: AI FOR SAFETY

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The call she is already pretending to be on, made real, with the alarm hidden inside the conversation.

18.61 min
national average 112 response time, 2025

72.3%
of women try to fake a phone call when travelling alone at night

95.6%
of accused in rape cases are already known to her

TEAM IDEAFORGE

Name	Institution	Role
Harsh Kawatra	Delhi Technological University (DTU)	Team Lead
Gursimran Kaur	GGSIPU	Member
Dayita Arora	Ramjas College, Delhi University (DU)	Member

WHAT PUKAAR IS

A two-way LLM voice call she starts before an unsafe commute.
It answers what she actually says, so it holds up to anyone listening.

One ordinary sentence sends her live location and never breaks the call.

PROBLEM STATEMENT

56% of Indian women are harassed on their commute. 52% then decline the job or the college seat that required it.

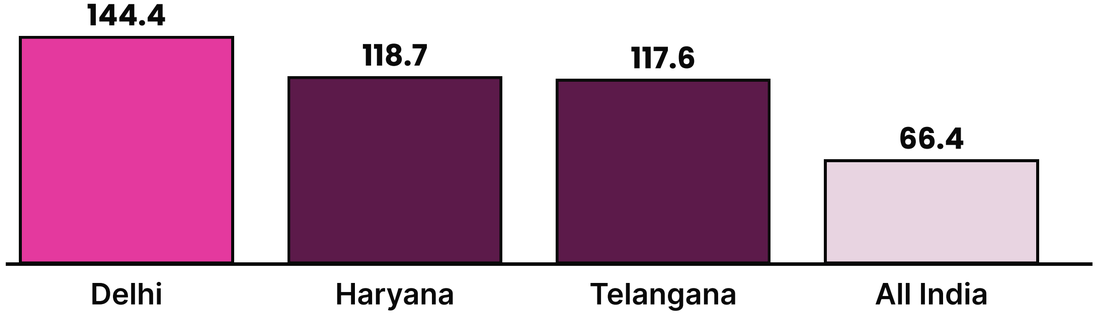
51
FIRs for crimes against women registered every hour, nationally

56%
of women report sexual harassment while commuting

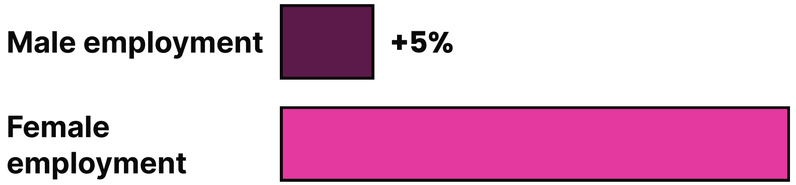
52%
declined an education or work opportunity over commute safety

Crimes against women per lakh female population

Delhi runs at 2.2 times the national rate.

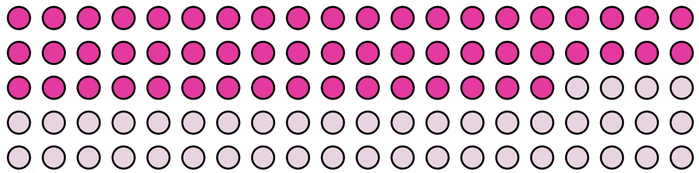


Delhi resettled 700,000 people to the periphery. Employment moved in opposite directions.



Her travel time tripled. The safety of the route decided whether she kept working.

Of every 100 women who commute



56 report harassment on the route

44 do not report it, which is not the same as safe

NFHS-5 puts lifetime spousal violence at 32%, so every count here is a floor.

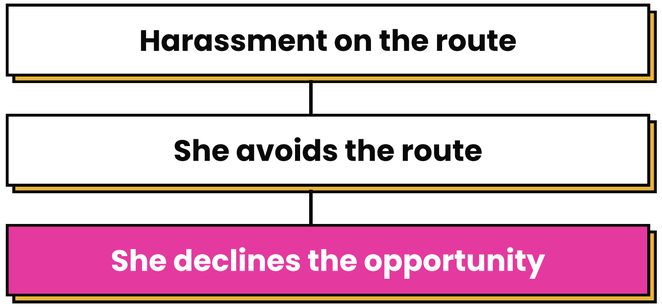
Who the accused actually is



Stranger, 4.4%

26,808 of 28,046 rape cases in 2020. The danger is usually someone she cannot visibly alarm without making things worse.

How a commute becomes a career decision



This never appears in any crime statistic.

Source: NCRB 2025 | NCRB 2020 | NFHS-5, 2019-21 | ORF 2021 | World Bank IEConnect South Asia | ILO 2017

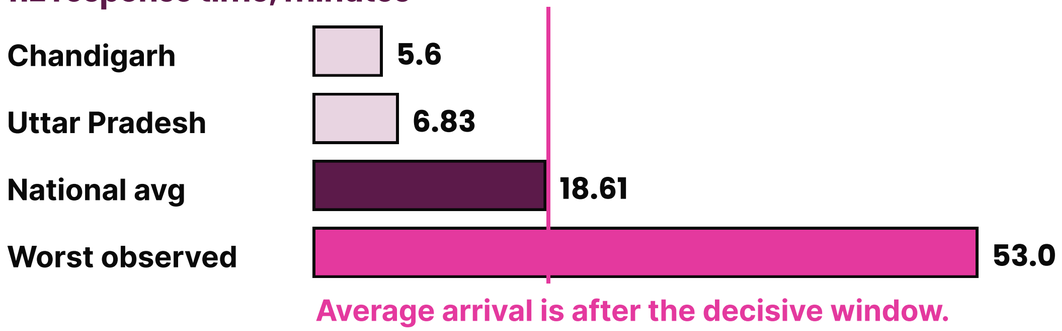
PROBLEM ANALYSIS

The alarm layer is mature. The missed window is before escalation, when she needs cover more than a button.

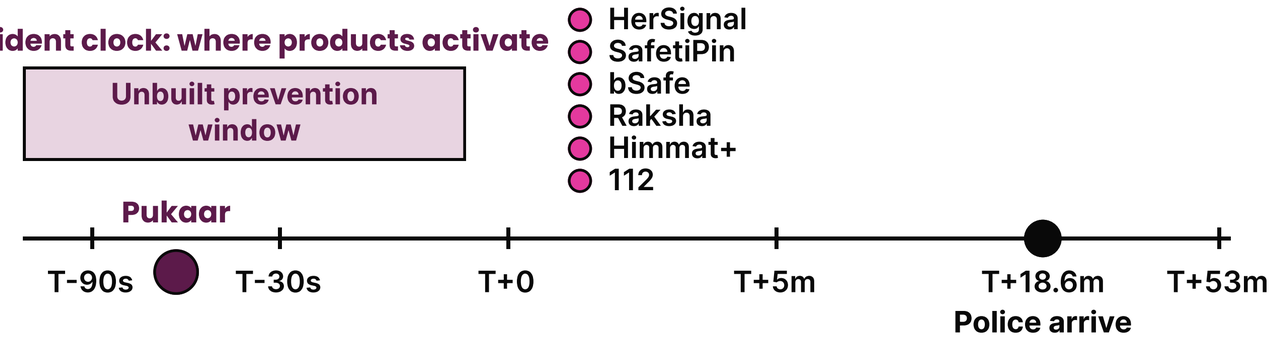
A decade of alarm-layer rollout



112 response time, minutes



Incident clock: where products activate



	112	Himmat	Raksha	bSafe	HerSignal	Pukaar
Trigger	Tap	Tap	SMS	Tap	Tap	Spoken
Watched phone	No	No	No	No	Partly	Yes
Two-way talk	No	No	No	No	No	Yes
Before act	No	No	No	Partly	Partly	Yes
India-first	Yes	Yes	Yes	No	No	Yes

HerSignal is closest, but its call is not a live two-way cover conversation.

Opportunity gaps

90 seconds before escalation

Works while the phone is watched

Turns an existing habit into product

Source: DoT panic-button rules | ERSS response data | rideshare safety survey, 2026 | product documentation

PROPOSED SOLUTION

Pukaar makes the pretend call real, then hides the alarm inside normal conversation.

1.2 s target first spoken word

2 independent alarm paths

0 visible actions at trigger time

From ordinary call to silent alert

1. **Starts session**
   A call screen opens before the walk.
2. **AI listens**
   It answers what she actually said.
3. **Duress line**
   A normal sentence carries intent.
4. **Alert sent**
   Known contacts receive live location.

DOES
Believable live conversation. Silent live-location alert.

DOES NOT DO
No idle locked-phone spying. No police call without her choice.

DEMO TRANSCRIPT

PUK Where are you right now, still near the market?

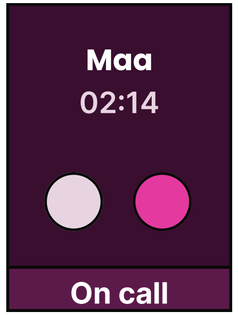
HER Just crossed it. Ten minutes, maybe.

PUK Okay, stay on the call. I will keep talking.

HER No, I already told mom. I will eat at home.

PUK Haha, fine. So listen, about tomorrow.

Alarm sent on line four. The call continues normally.



No safety signal on screen.

Two alarm paths

Path 1
Model calls tool

Path 2
Local phrase match

RAISE ALARM

Path 2 still fires if relay is offline.

Source: Web Speech API | MediaDevices specification | Pukaar system design

TECHNICAL ARCHITECTURE

Two independent trigger paths keep the alarm alive even when the voice relay fails.

Signal path

PWA client	Voice relay	STT	LLM persona	TTS	Dispatcher	Contacts
Next.js PWA	Cloud Run	partials	raise alarm tool	audio back	Cloud Run	FCM, WA, map

Alarm branches at the LLM and does not wait for speech playback.

Latency budget to first word



- Mic 120 ms
- STT 220 ms
- LLM 380 ms
- TTS 300 ms
- Play 180 ms

Two tiers, one contract

Tier A	Streaming STT plus LLM plus TTS
Tier B	Realtime speech-to-speech demo path

Both expose raise alarm system.

Layer	Choice	Reason
Client	Next.js PWA	Installable
Voice	STT, LLM, TTS	Low cost
Trigger	Tool plus local match	Redundant
Dispatch	Firestore, FCM, WA	Demo-safe
Host	Vercel and GCP	Billing ready

PRIVACY BOUNDARY

Never stored	Session only
Audio	Session id
Transcripts	Contacts
Nearby data	Live trail
	Duress time

TTL clears trail after session.

DATA FLOW DETAIL

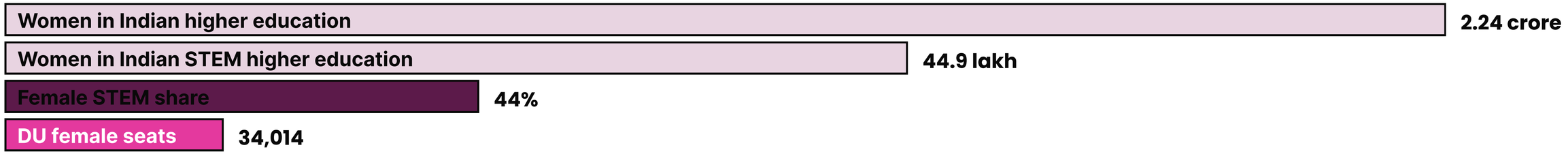
- Mic capture streams 16 kHz PCM to Cloud Run.
- Local phrase matcher runs in browser.
- LLM can call raise_alarm silently.
- Firestore stores only session state.
- Contact link opens live map with token.
- No audio or transcript persistence.

Source: Vercel, Google Cloud and OpenAI docs | DPDP Act 2023 | Pukaar system design

EXPECTED IMPACT

Pukaar starts with 34,014 DU women admitted in one round, with Tier A voice cost near Rs 39 per protected commute.

Named-count reach funnel



Cost line	Tier A	Tier B
STT per min	$0.016	included
Model per min	~$0.00025	token priced
TTS per min	~$0.004	included
20-min commute	~$0.405	usage varies
Protected commute	~Rs 39	premium path

NOW

Prototype on one campus. Hinglish voice. Two triggers live.

NEXT

Regional voices. Group contact fan-out. Battery-aware sessions.

LATER

Transport integration. Offline fallback needing no data.

Risk	Why it matters	Mitigation
False trigger	Accidental phrase	Dedupe plus silent cancel
Cost	Voice is not free	Tier A default
Latency	Stutter breaks cover	Local filler lines
Surveillance misuse	Location can be coerced	User controls contacts

Future scope



Source: AISHE 2023-24, PIB | DU admissions reporting, 2025 | Google Cloud and OpenAI pricing, accessed 30 Aug 2026

Thank you

Pukaar. A call, and a cry for help.

Team IdeaForge

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